BACON BROS’ Design Doc for ***P04***

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PROJECT NAME: Lettuce Relax

TARGET SHIP DATE: 2026-4-20

Overview:

The initial load page will introduce you to food consumption and exercise worldwide through statistics, focusing on the health of all populations (debating whether or not to limit scope to youth or include everyone). The surprise is the interactive part where the visitor will be able to learn what food will provide them with what caloric intake, and what exercises they can do to burn off said calories (based on their traits?). The site will allow them to craft their own personalized dietary and workout goals, the progression of which will also be displayed with data visualization tools listed below.

Program Components

sqlite3:
user_info

Javascript:
chart.js

Flask:
__init__.py

HTML:
Log in + Sign up
Home
Explore page featuring graphs
Profile
Personalize page featuring personalized projections

FEF:
Bootstrap

Component Explanation:

sqlite3: used to store user info, including account information and tailored dietary/exercise plans.

chart.js is the javascript folder in which we will be able to create functions that take in parameters such as calories, nutrients, etc. and convert them into appropriate polar area, bar, and line graphs. We chose chart.js because of its aesthetic appeal and because of the helpful documentation/examples.

init.py is the main flask app handler, persistently hosted by nginx

bootstrap is how we will make the website interactive and visually appealing

The ***user_info*** database will be used to add/remove displayed information using data (match user to a similar lifestyle) in the other dbs.

API none planned for now

data.py Turns data file into a SQLite table by skimming through our CSV files using a python script. This script opens the CSV files one by one using a for-loop, and uses SQLite to transfer the necessary information into one .db file which will contain the table of data later used by our JS. This function should happen once so that only the data we use/need will be parsed through later on. Python file has code that retrieves specific data rows from that .db table, which will later be used to view specific data points for comparison or to create a dropdown menu based on user search input.

Datasets:

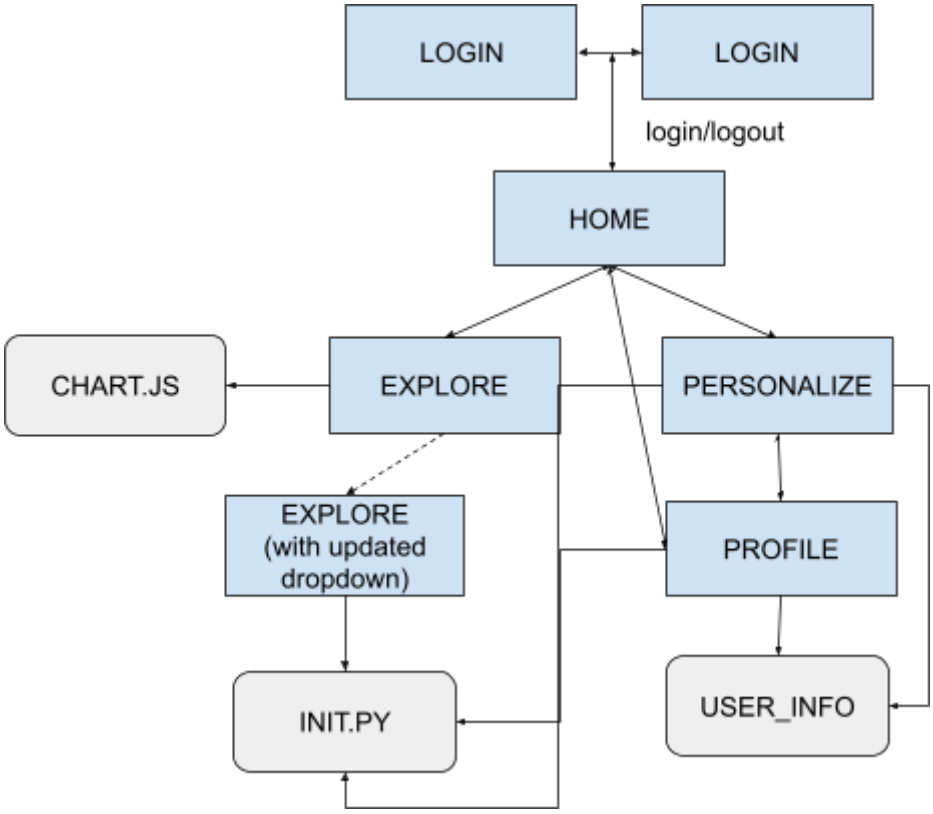
- FOOD NUTRITION: https://www.kaggle.com/datasets/utsavdey1410/food-nutrition-dataset (to provide data regarding foods and their nutrition, calories, etc)
- WORKOUTS: https://www.kaggle.com/datasets/jockeroika/life-style-data (to provide information about workout times, instructions, and calories burnt)

Dataset Display Methods:

- **POLAR AREA:** https://www.chartjs.org/docs/latest/samples/other-charts/polar-area.html (to allow visual comparisons of individual foods' stats such as nutrients, calories, vitamins, etc)
- **BAR GRAPH:** https://www.chartjs.org/docs/latest/samples/bar/border-radius.html (for easy comparison between nutritional value of different dishes)

- **Line Chart:** https://www.chartjs.org/docs/latest/samples/line/line.html (display long-term projections set by personalized plan)

Component Map:



Database Organization

USER_INFO		
TEXT	user	PK
TEXT	password	NOT NULL
TEXT	dietary_restrictions	
TEXT	exercises	
TEXT	goals	

The user and password will be created by the user upon sign up, where both can be edited through edit_info page. Information on dietary_restrictions and goals will also be edited through edit_info and affects your recommended exercises. Exercises will be recommended by program based off of given information.

Datasets (will contain the data from their respective csv files):

Food_Nutrition		
INT	item	PK
TEXT	name	
REAL	calories	
REAL	sugar	
REAL	protein	
REAL	fiber	
TEXT	fats	
REAL	water	
REAL	cholesterol	
TEXT	vitamins	
TEXT	minerals	

Fast_Food		
INT	item	PK
TEXT	restaurant	
TEXT	name	
INT	calories	
INT	sugar	
INT	protein	
INT	fiber	

TEXT	fats	
INT	cholesterol	
TEXT	vitamins	
TEXT	minerals	

Workout		
INT	life	PK
TEXT	type	
INT	age	
TEXT	gender	
REAL	weight	
REAL	height	
REAL	BPMe	
REAL	BPMr	
REAL	duration	
REAL	calsBurned	
REAL	waterIntake	
INT	freq	
REAL	experience	

Site Map (front end):

Team Roles:

Veronika(PM): CSS, home page, personalize page, algorithm

Andrew(scribe): FEF, explore page, chart.js, dropdown

Alvin: home page, explore page, chart.js, data parsing, storage

Naomi: registration, login, health data storage, algorithm, profile page